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YUWON KIM

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VIRTUAL REALITY · MULTIMODAL ANALYSIS · HUMAN-COMPUTER INTERACTION

A Human-Computer Interaction (HCI) researcher specializing in *Virtual Reality*, *multimodal behavioral sensing*, and *explainable AI* to turn everyday human behavior into insight that supports better decisions. **Analyzing behavioral data captured during everyday activities and interactions to generate interpretable, actionable insights that support human decision-making** — spanning eye movement, hand movement, and physiological signals sensed through VR and wearable devices. Developed and validated this approach by building VR-based digital biomarker systems for the early screening of mild cognitive impairment, through multidisciplinary collaboration with clinical experts (Asan Medical Center, Seoul), engineers at a national research institute (KITECH), and international academic partners (Zurich University of Applied Sciences). Now extending this foundation toward embodied and trusted human-AI collaboration contexts — designing systems that sense, interpret, and support human decisions wherever they occur.

Education

PhD Candidate, Department of Applied Artificial Intelligence — 2022.03.01. ~ 2027.02.23. (expected)
Seoul National University of Science and Technology, Seoul, Korea

- Advisor: Prof. Kyoungwon Seo
- Thesis: (in progress) An Information Fusion Framework of VR-Based Digital Biomarkers for Early Screening of Mild Cognitive Impairment
- GPA: 4.18 / 4.5

BS, ICT Artificial Intelligence (Interdisciplinary Convergence Double Major) — 2020.03.16. ~ 2022.02.17.
Seoul National University of Science and Technology, Seoul, Korea

- Advisor: Prof. Kyoungwon Seo, Prof. Ji-Hyeong Han
- Thesis: A Study on the Cognitive Load in Stressful Situations Through Eye Tracking Analysis in Virtual Reality Kiosk Ordering Task

BS, Optometry (Primary Major) — 2016.03.02. ~ 2022.02.17.
Seoul National University of Science and Technology, Seoul, Korea

- Advisor: Prof. Hyeran Noh
- Thesis: Malaprade Reaction–Based Enhancement of Color Vividness in Paper Sensors and Proposal of Efficient Conditions for a Target Shrinkage Ratio
- Total GPA: 3.5 / 4.5

Journal Articles (More info: Google Scholar)

Click any entry below to view the full paper.
† = Equal Contribution;

[J4] Park, B.† , **Kim, Y.**† , Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024). Integrating biomarkers from virtual reality and magnetic resonance imaging for the early detection of mild cognitive impairment using a multimodal learning approach: validation study. *Journal of medical Internet research, 26*, e54538. **[SCIE, Q1, Top 3%]**

[J3] Kim, D., **Kim, Y.**, Park, J., Choi, H., Ryu, H., Loeser, M., & Seo, K. (2024). Exploring the relationship between behavioral and neurological impairments due to mild cognitive impairment: Correlation study between virtual kiosk test and EEG-SSVEP. *Sensors, 24*(11), 3543. **[SCIE, Q1]**

[J2] **Kim, S. Y.**, Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2023). Digital marker for early screening of mild cognitive impairment through hand and eye movement analysis in virtual reality using machine learning: First validation study. *Journal of medical Internet research, 25*, e48093. **[SCIE, Q1, Top 3%]**

[J1] Kim, S. Y., Park, H., Kim, H., Kim, J., & Seo, K. (2022). Technostress causes cognitive overload in high-stress people: Eye tracking analysis in a virtual kiosk test. *Information Processing & Management*, 59(6), 103093. **[SSCI, Q1, Top 5%]**

Conference Papers (More info: Google Scholar)

Click any entry below to view the full paper

[C9] Kim, Y. (2025, April). Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems* (Student Research Competition). **[Research Competition Second-Place Award; acceptance rate: 15%]**

[C8] You, D., **Kim, Y**., & Seo, K. (2025, April). Tailored Virtual Agent Guidance for Stress Management: Comparing Directive and Non-Directive Approaches by Alexithymia Status. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 32.8%]**

[C7] Cha, S. E., **Kim, Y**., & Seo, K. (2025). Effects of Tool Description Language on Language Model's Tool Calling in Korean Dialogue. In *Proceedings of the 2025 HCI Korea Conference*.

[C6] You, D., **Kim, Y**., & Seo, K. (2025). Digital Mental Health Intervention with Virtual Agent: Impact of Guidance Approaches on Stress Reduction in Young Adults. In *Proceedings of the 2025 HCI Korea Conference*.

[C5] Keum, H., **Kim, Y**., You, D., & Seo, K. (2025). A Deep Learning Model for Stress Level Recognition Utilizing Neutral and Stress State of Remote Photoplethysmography. In *Proceedings of the 2025 HCI Korea Conference*.

[C4] Kim, Y., Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2024, May). Decoding Behavior: Utilizing Virtual Reality Digital Marker and Machine Learning for Early Detection of Mild Cognitive Impairment. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 33.8%]**

[C3] Park, B., **Kim, Y**., Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024, May). Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 33.8%]**

[C2] Kim, S. Y., Park, B., Kim, D., Choi, H., Park, J., Ryu, H., & Seo, K. (2024). Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning. In *2024 International Conference on Electronics, Information, and Communication (ICEIC)*.

[C1] Kim, S. Y., Park, J., Choi, H., Ryu, H., & Seo, K. (2023). Virtual Kiosk Test for Early Screening of Mild Cognitive Impairment through Eye Tracking Data Analysis. In *Proceedings of the 2023 HCI Korea Conference*.

Patents

[P2] Seo, K., **Kim, S. Y.** (2025). VR(Virtual Reality) Digital Biomarker based Clinical Decision Support System and Method using Artificial Intelligence for Early Diagnosis of Alzheimer's Disease. *KR10-2025-0022344*. (published 2025.02.17)

[P1] Seo, K., **Kim, S. Y.** (2024). Eye tracking data calibration method and eye tracking data calibration system. *KR10-2024-0048131*. (published 2024.04.15)

Research Grants

NRF, Research Subsidies for Ph.D. Candidates (PI) 25,000k KRW 2025.09.01. ~ 2026.08.31.

- Title: LLM-Based Clinical Decision Support Tool for Alzheimer's Disease via Multimodal Reasoning Integrating VR, MRI, and Neuropsychological Tests

Awards and Honors

Seoul AI Tech Graduate Scholarship (Seoul Scholarship Foundation, 10,000k KRW) 2025.
Research Competition Second-Place Award (ACM CHI 2025) 2025.

Teaching

Deep Learning (Assistant) Fall 2022

- Learn deep neural network principles, mathematical foundations, and key models including autoencoders and GANs
- Conduct a term project applying deep learning models to address real-world problems

Selected Talks & Professional Service

Workshop Organization

- Co-Facilitator, *Augmented Educators and AI: Shaping the Future of Human-AI Collaboration in Learning*, ACM CHI 2025

Peer Review

- Reviewer, ACM CHI Conference on Human Factors in Computing Systems (CHI 2026)
- Reviewer, ACM Transactions on Computing for Healthcare (HEALTH)

Technical Skills

- Programming: Python
- VR/AR/Spatial Computing Development: Unity; Apple Vision Pro app development

References

Prof. Kyoungwon Seo
Professor, Dept. of Applied Artificial Intelligence, Seoul National University of Science and Technology, Seoul, Korea
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Prof. Sidney Fels
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Prof. Seong-Eun Kim
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